

Question:

A 34-year-old woman presents to her OB/GYN with complaints of missing her last 3 periods as well as intermittent spontaneous milky-white nipple discharge bilaterally for the past 3 months. Vital signs are stable and within normal limits. Neurologic examination is without abnormality, including normal visual fields. Serology and MRI of the brain are ordered, with results pending. Which of the following sets of laboratory results would be expected in this patient?

Answer Choices:

- A. Decreased prolactin, decreased FSH, decreased LH
- B. Decreased prolactin, increased FSH, increased LH
- C. Increased prolactin, decreased FSH, increased LH
- D. Increased prolactin, decreased FSH, decreased LH

Reasoning + Answer:

<think>

Finding Reasoning Paths :

1. **Prolactinoma (pituitary tumor)** : A prolactin-secreting tumor could lead to high prolactin levels , which may suppress the hypothalamic-pituitary axis , causing decreased levels of FSH and LH .
2. **Hypothalamic dysfunction** : Conditions affecting the hypothalamus , such as rare diseases or structural abnormalities , could disrupt the release of releasing hormones , leading to decreased FSH and LH production .
3. **Polycystic ovary syndrome (PCOS)** : This condition can cause ovarian dysfunction , leading to decreased FSH and LH secretion .
4. **Medications (e.g. , metoclopramide or domperidone)** : These drugs can increase prolactin levels , potentially causing decreased FSH and LH .
5. **Secondary amenorrhea** : Hormonal imbalances could be secondary to a pituitary or hypothalamic cause , leading to decreased FSH and LH .

Reasoning Process :

1. **Prolactinoma (pituitary tumor)** :
 - Prolactinomas are the most common type of pituitary adenoma , and they are often microadenomas .
 - A prolactin-secreting tumor can cause hyperprolactinemia , which suppresses gonadotropin-releasing hormone (GnRH) release from the hypothalamus .
 - This suppression leads to decreased secretion of FSH and LH from the anterior pituitary .
 - The milk production and abnormal nipple discharge are consistent with this condition , as hyperprolactinemia can lead to galactorrhea and ovarian dysfunction .
2. **Hypothalamic dysfunction** :
 - Conditions affecting the hypothalamus , such as genetic syndromes (e.g. , McCune-Albright syndrome) , can disrupt the release of GnRH .
 - This would result in decreased FSH and LH secretion , potentially leading to amenorrhea and other hormonal imbalances .
3. **Polycystic ovary syndrome (PCOS)** :
 - While PCOS is a common cause of hyperandrogenemia and infertility , it is less likely to be the cause in this case , as the patient has missed her last three periods and has not reported symptoms like acne or hirsutism .
4. **Medications (e.g. , metoclopramide or domperidone)** :
 - These drugs are known to increase prolactin levels by binding to dopamine receptors in the pituitary gland .
 - Elevated prolactin levels can suppress GnRH release and lead to decreased FSH and LH production .
5. **Secondary amenorrhea** :
 - This condition refers to the absence of menstruation in the absence of pregnancy , and it could be secondary to a pituitary or hypothalamic cause .
 - The combination of decreased FSH and LH is a hallmark of hypogonadotropic hypogonadism , which could be caused by a pituitary adenoma or hypothalamic dysfunction .

Conclusion :

The most likely cause of the decreased FSH and LH , along with the absence of menses and milk production , is **hyperprolactinemia due to a prolactin-secreting pituitary adenoma (prolactinoma)** . This condition explains the galactorrhea , amenorrhea , and decreased gonadotropin levels .

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<answer>

Increased prolactin , decreased FSH , decreased LH

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